The Snowplow Problem

Sierra & Rachel

MATH-345: Differential Equations

March 31, 2026

Table of Contents

Section 1

Chapter Two Fundamentals

Context from Chapter 2

- The snowplow problem is models a physical process where the rate of change of a dependent variable (x, the distance the snowplow travels), depends on an independent variable (time, t) and another function (snow depth h), which increases over time.
- We utilize multiple first-order, linear ODEs
- These are solved using separable equations, initial conditions, and integrating factors

Section 2

One Plow Problem

Problem Statement

Fundamentals of Differential Equations, pg 84:

> *One day it started snowing at a heavy and steady rate. A snowplow started out at noon, going 2 miles the first hour and 1 mile the second hour. What time did it start snowing?*

Preliminary Assumptions

1. Because it is snowing steadily, we can assume it is a *constant rate*
2. The deeper the snow, the slower the plow, implying that the rate the snow plow moves is *inversely proportional* to the depth of the snow

Variables & Notation

- t: the time after noon (hours)
- $x(t)$: distance the plow traveled (miles)
- $h(t)$: height of the snow (inches)
- b: hours before noon the snow started (hours)
- k_1: rate of the snowfall (inches/hour)

Further Assumptions

From the variables, we can also assume

- Given $h(t)$ is the height and k_1 is the height over time, $h'(t) = k_1$
- This implies that $h(t) = k_1 t + C$
- Given $x(t)$ is the distance the plow has traveled, $x'(t)$ will be the speed of the plow.

Section 3

Solving One Plow

Height of Snow

Given our assumption that $h(t) = k_1 t + C$, we can solve for C my inffering at $-b$ hours before noon, there was no snow, $h(t) = 0$. This provides the initial conditions $h(-b) = 0$,

$$h(-b) = k_1(-b) + C = 0,$$

suggesting $C = k_1 b$. Therefore

$$h(t) = k_1 t + k_1 b = k_1(t + b).$$

Snowplow Speed

Assume snowplow speed is inversely proportional to the height of the snow. We know $x'(t)$ is the speed of the plow, this suggests $x'(t)h(t) = k_1$. Therefore,

$$x'(t) = \frac{k_2}{h(t)} = \frac{k_2}{k_1(t+b)} = \frac{k_3}{(t+b)},$$

where $k_3 = \frac{k_2}{k_1}$. Now, integrating both sides results in

$$x(t) = k_3 \ln(t+b) + C_1.$$

First Initial Conditions

Given the condition that the plow traveled 2 miles in the first hour, $x(1) = 2$. This suggests

$$x(1) = k_3 \ln(1 + b) + C_1 = 2.$$

We let the constant value $C_1 = -k_3 \ln b$. Then,

$$\begin{aligned} x(1) &= 2 \\ k_3(\ln(1 + b) - \ln(b)) &= 2 \\ k_3 \ln \frac{(1 + b)}{b} &= 2 \end{aligned}$$

Second Initial Conditions

Given the condition that the plow traveled 1 mile in the second hour, $x(2) - x(1) = 1$. Therefore,

$$\begin{aligned} x(2) - x(1) &= 1 \\ k_3 \ln(2+b) - k_3 \ln(1+b) &= 1 \\ k_3 \ln\left(\frac{2+b}{1+b}\right) &= 1. \end{aligned}$$

We multiply both sides by 2 to get

$$\begin{aligned} 2k_3 \ln\left(\frac{2+b}{1+b}\right) &= 2 \\ k_3 \ln\left(\frac{2+b}{1+b}\right)^2 &= 2. \end{aligned}$$

Solving the IVP

Now, we have two functions that equal 2,

$$k_3 \ln\left(\frac{2+b}{1+b}\right)^2 = 2$$

$$k_3 \ln\frac{(1+b)}{b} = 2$$

So we set the functions equal

$$k_3 \ln\left(\frac{2+b}{1+b}\right)^2 = k_3 \ln\frac{(1+b)}{b}$$

$$\ln\left(\frac{2+b}{1+b}\right)^2 = \ln\frac{(1+b)}{b}$$

More IVP Solving

We then can assume that

$$\left(\frac{2+b}{1+b}\right)^2 = \frac{(1+b)}{b}$$

So, with cross multiplication,

$$\begin{aligned} (1+b)^3 &= b(2+b)^2 \\ 1+3b+3b^2+b^3 &= 4b+4b^2b^3 \\ b^2+b-1 &= 0 \\ b = \frac{-1\pm\sqrt{5}}{2} & \end{aligned}$$

Final Solution

We know that since b is the time before noon, we assume $b > 0$, so $\frac{-1+\sqrt{5}}{2} \approx 0.618$.

Therefore, the snow started 0.618 minutes before noon, or 37 minutes before.

The snow started at 11:23 A.M.

Section 4

Application with Two Plows

Fundamentals of Differential Equations, pg 84:

> *One day, it began to snow exactly at noon at a heavy and steady rate. A snowplow left its garage at 1:00 p.m., and another one followed in its tracks at 2:00 p.m. At what time did the second snowplow crash into the first?*

Two Plows Figure

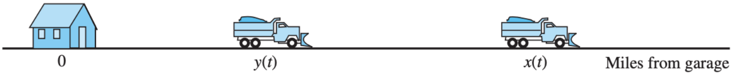

Figure: Figure 2.15

2 Plow Problem Set-up

Snowplow 1 ($t \geq 1$)

$$\frac{dx}{dt_1} = \frac{k}{h_1} = \frac{k}{H(t_1 + T)}$$

Snowplow 2 ($t \geq 2$)

$$\frac{dx}{dt_2} = \frac{k}{h_2} = \frac{k}{H(t_2 + T)}.$$

More Set-Up

We invert the two equations and let $A = \frac{k}{H}$,

$$\frac{dt_1}{dx} = A(t_1)$$

$$\frac{dt_2}{dx} = A(t_2 - t_1)$$

where the initial conditions for t_1 being 1:00 PM and t_2 bring 2:00 PM are

$$t_1(0) = 1 \text{ and } t_2(0) = 2.$$

How to Solve

We rearrange our first-order ODEs to

$$\frac{dt_1}{t_1} = Adx$$

$$\frac{dt_2}{dx} - At_2 = -At_1$$

where we will replace t_1 in the second equation after solving the first.

Solving for $t_1(x)$ with Separable Equation

First ODE equation solving for t_1

$$\frac{dt_1}{t_1} = A dx,$$

Integrating results in

$$\begin{aligned} \ln t_1 &= Ax + C \\ t_1(x) &= e^{Ax+C} \\ t_1(x) &= C_1 e^{A_x}. \end{aligned}$$

Given the initial conditions $t_1(0) = 1$,

$$\boxed{t_1(x) = e^{Ax}}$$

Solving for t_2 with integrating factor

Begin with substitution

$$\frac{dt_2}{dx} - At_2 = -At_1$$
$$\frac{dt_2}{dx} - At_2 = -Ae^{Ax}.$$

We find the integrating factor

$$P(x) = e^{\int(-Adx)} = e^{-Ax}.$$

Multiplying and rewriting results in

$$\frac{d}{dx}(e^{-Ax}t_2) = -Ae^{Ax}e^{-Ax} = -A.$$

SOLVING t_2 CONT.

Integrating with respect to x on both sides

$$e^{-Ax} t_2 = -Ax + C_2,$$

and solving for t_2,

$$t_2(x) = -Axe^{Ax} + C_2 e^{Ax}.$$

The initial conditions imply that

$$t_2(0) = -A(0)e^{A(0)} + C_2 e^{A(0)} = 2 \implies C_2 = 2.$$

Thus,

$$\boxed{t_2(x) = e^{Ax}(2 - A).}$$

Solving Crash

When the two snowplows collide, $t_1(x)$ and $t_2(x)$ must exist at the same point, X_C, at the same time, T_C. This gives us another system of equations!

$$t_1(X_C) = T_C$$
$$t_2(X_C) = T_C$$

Solving T_C

Using solved equations from above

$$t_1(X_C) = e^{AX_C} = T_C$$
$$t_2(X_C) = e^{AX_C}(2 - A) = T_C$$

In order to solve, let

$$e^{AX_C} = e^{AX_C}(2 - A)$$

Dividing by e^{AX_C} results in

$$1 = 2 - AX_C \implies AX_C = 1.$$

Finishing Crash Solution

Given $AX_C = 1$,

$$t_1(x) = e^1.$$

This means that $t_1(x) = t_2(x)$ around $e \approx 2.718$ hours after noon. So, the two plows collided at about 2:43 P.M.

Section 5

Conclusions

Conclusions

This problem was solved by using first-order, linear, differential equations. We could have also solved with modeling simulations and trigonometry concepts.

Citations

"Snow plow problem." Oregon State University.

Bender & Orszag Mathematical Methods: Chapter 1 - Problem 1.22

R. K. Nagle, E. B. Saff, and A. D. Snider, in *Fundamentals of Differential* Equations, 2018, p. 84.

R. Wright, "My favorite math problem." April 2016.